TECHNICAL MEMORANDUM / EVIDENCE RECOVERED

Memory on physical silicon

Recorded FLASH, RAM and stack observations across ESP32-S3, STM32F446RE and CH32V307. The figures below are preserved measurements from different firmware artifacts - not an architecture-wide memory claim.

100,000	67.89 KB	86,124 B
ESP32-S3 Step20 cycles	ARM 100K flashed payload	RISC-V R1 only; not V2 100K

Comparative evidence register

TARGET & ARTIFACT	FLASH / IMAGE RECORD	RAM EVIDENCE	STACK EVIDENCE
ESP32-S3 Step20 endurance 17-18 Sep 2026	287,672 B Sum of six boot-log app segments; not a standalone ELF text+data report.	368,691 B internal heap free; 16,774,624 B PSRAM free at the 100K checkpoint.	440 B FreeRTOS stack high-water mark: minimum unused task stack at 100K.
STM32F446RE Cortex-M4 100K 03 Oct 2026	67.89 KB programmed image reported by STM32CubeProgrammer for osh_stm32_100k.elf.	Matched text/data/bss for the 100K image: not established.	Peak stack consumption: not recorded for the exact 100K image.
CH32V307VCT6 DILIGENCE_R1 only Not V2 100K	86,124 B R1 loadable firmware image (NOT the V2 100K build); local HEX byte count matches.	5,484 B static RAM = 152 B .data + 5,332 B .bss.	2,048 B stack section reserved; measured peak use not available.

Reading rule. These are whole-firmware figures (startup, drivers and libraries included), so for the tested native path they are upper bounds. This statement applies to image/payload footprints, not the separate free-heap or stack observations. Different measurement boundaries cannot be treated as one complete FARO footprint.

A smaller experimental ARM ELF previously reported as 1,124 B FLASH / 1,988 B static RAM is excluded from this table: it is not the proven 100K image and does not describe the whole FARO architecture.

FIRMWARE IDs / WHAT THEY DO AND DO NOT PROVE

Binary identity and toolchain

01 / STM32F446RE (ARM Cortex-M4)

Physical 100K record: `osh_stm32_100k.elf`; 100,000 cycles, 0 ADC failures, 0 state failures; STM32CubeProgrammer reported 67.89 KB programmed. [S3]

Archive boundary: The 100K ELF name is present in the physical migration record, but its original ELF bytes are not available here. An independently matched SHA-256 for that named path was not established from accessible primary records.

Separately named artifact: `osh_stm32_uart.elf`, identified by SHA-256 in the ARM validation report [S4]. Whether it matches the named 100K image is unresolved.

Recorded ELF SHA-256 (osh_stm32_uart.elf)

```
9cd4c56ab5c29bb150900c96abff3328
fdeb1b28e13b639fd97fbb0f6a7a0645
```

> Identity boundary: The 100K programming record names one ELF and the contemporaneous hash table names a UART ELF. The available documents alone do not prove they are identical or different. Do not assign the UART hash or unmatched memory figures to the 100K build. Original ELF bytes were unavailable for a fresh `size` measurement.

Build context: a related 06 Oct 2026 ARM command uses GCC for Cortex-M4, `-0s`, section garbage collection, CMSIS headers, `-specs=nosys.specs` and `-lm`. It is not a verified build command for the 100K ELF. The linked code is not proven libc-free or HAL-free. [S6]

02 / CH32V307VCT6 (RISC-V)

R1 only (not the 100K V2 build): `OSH_RISCV_DILIGENCE_R1.elf`. Section dump: `.text=0x14e14`, `.data=0x98`, `.bss=0x14d4`, `.stack=0x800`. R1 loadable image: 86,124 B; matching local HEX byte count: 86,124. [S2, S5]

Locally checked HEX SHA-256 (OSH_RISCV_DILIGENCE_R1.hex)

```
52a8e10666c500ce84580a32d1231917
c3b5afe1ee013e1cf00504e7056bc7e7
```

Separate 100K V2 record: the physical endurance V2 ELF has SHA-256 below, but its text/data/bss footprint was not recovered. The R1 memory numbers above must not be assigned to the V2 100K firmware. [S7]

RISC-V V2 100K ELF SHA-256

```
cb9f88f38e755cb82f51fdfe0bf8000f
f0963eb774195e1b3b965ae1975c1517
```

03 / ESP32-S3 (Xtensa LX7)

Measured runtime: Step20 100K completed in 47.112 s with 0 cognitive failures. The 17 Sep 2026 firmware boot identifies project `OSH_ESP32_STEP20_STABILITY_V1_F`, ESP-IDF v5.4.4, with ELF hash shown only as the truncated prefix `41bbe7034...`. A complete hash for this precise booted ELF was not recovered. [S1, S8]

The 287,672 B figure sums six app segments from the boot log. It is not static RAM use or partition size. Free heap and task-stack HWM were recorded through 100K cycles. The full Step20 ELF SHA-256 needs the original ELF bytes. [S1, S8]

PUBLIC DISCLOSURE / STRICT SCOPE

How to publish these figures

Measurement definitions

FLASH image. For GCC ELF targets, the conventional approximation is .text + .data, but firmware also includes startup, vector tables, read-only sections, alignment and other loadable regions. A programmer-reported payload or sum of boot segments is not automatically an equivalent "flash usage" measurement.

Static RAM. Usually .data + .bss (and other writable allocated sections as applicable). This excludes dynamic heap allocation and maximum active stack use.

Stack. A reserved .stack section is capacity, not observed peak consumption. FreeRTOS stack HWM is the minimum stack space ever left unused; calculating peak requires the allocated task-stack size.

FARO boundary. The records validate specified native numerical, causal or embedded paths; they do not establish that every FARO component, including the full language system, runs in the measured MCU images.

Website-ready wording (English)

> Embedded memory evidence. The image-size figures cover whole firmware, including startup, drivers and linked libraries; they are upper bounds for the tested native path, not a measured whole-FARO installation. OSH maintains physical memory records for ESP32-S3, STM32F446RE and CH32V307. On ESP32-S3, the Step20 endurance test retained 368,691 B of free internal heap and 16,774,624 B of free PSRAM at 100,000 cycles, with a 440 B task-stack high-water mark (unused stack). On CH32V307, a preserved R1 firmware image contains 86,124 B of loadable payload and 5,484 B of static RAM, with a separate 2,048 B reserved stack. On STM32F446RE, a 100K validation image was physically programmed and verified, with 67.89 KB reported by the programmer. These are artifact-specific measurements obtained by different methods, not a complete FARO architectural memory footprint. Firmware identity and stack/heap limits are disclosed in the accompanying evidence record.

Original source register

[S1] OSH_ESP32S3_EVIDENCIA_INTEGRADA_2026-09-18.pdf, pp. 6-7 (Step20 / memory, stack, 100K).

[S2] OSH_R1_ARRANQUE_COMPILADO.txt (ELF headers, sections and stack reservation, 08 Oct 2026).

[S3] OSH_Roadmap_ARM_Physical_Migration_Completed.pdf, pp. 1-2 (programmed ARM image and physical 100K); no verified SHA-256 of the named programmed ELF available here.

[S4] ARM physical validation report, 3 Oct 2026 (p. 7; distinct UART ELF/BIN hashes).

[S5] HEX_LOCAL_VERIFICADO.json (local RISC-V HEX payload and file hash; no device readback).

[S6] Texto pegado(20261006-202311).txt (related ARM build command and linker warnings).

[S7] OSH_RISCV_Registro_Tecnico_Validacion_Fisica_2026-10-08.pdf, pp. 2-4 (V1/V2 identities, 20K/100K).

[S8] Texto pegado(8).txt (ESP32-S3 boot segments, task memory telemetry and truncated ELF hash).

Evidence status: historical records reconciled on 04 Nov 2026. The ARM 100K, RISC-V V2 100K and ESP32-S3 Step20 ELF files were not available for new size/hash computations. No new hardware measurements were made. Proprietary source is not disclosed.